Nazar Budaiev

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Education

Ph.D. in Astronomy, University of Florida *May 2026*
Dissertation: Extreme Star Formation in Sagittarius B2. Advisor: Adam Ginsburg

M.S. in Astronomy, University of Florida *August 2022*

B.A. in Astronomy & Physics, Boston University *May 2020*

Research Experience

Postdoctoral Associate *2026 – present*
University of Florida

- Co-lead data reduction and analysis for a JWST large program surveying the Galactic center (PID: 10678).
- Designing and maintaining a pipeline that improves astrometric accuracy and source extraction in complex fields.
- Developed a pipeline to utilize MIRI coronagraph panes for imaging purposes, increasing survey's coverage by 20%.

Graduate Research Assistant *2020 – 2026*
University of Florida

- Investigated star formation in the Galactic center with JWST, ALMA, and VLA; primary focus on Sagittarius B2.
- Evaluated and re-imaged data as part of data reduction team for ACES ALMA large program, coordinating with an international team of >100 investigators across multiple institutions.
- Contributed to 16 accepted telescope proposals across JWST, ALMA, GBT, and VLA.

Visiting Researcher *Spring 2025*
University of Connecticut

- Advised on source extraction and classification strategies for the ACES large ALMA program.
- Analyzed and characterized star-forming populations in the Galactic center using the ACES data.

Research Assistant *2018 – 2020*
Boston University

- Characterized GBT's VEGAS receiver; modeled HI clouds in the Milky Way with IDL.

Summer Research Intern *Summer 2019*
Green Bank Observatory

- Created a 6 GHz continuum map of the inner Galactic Plane using GBT data in IDL and Python.

Selected Talks

- Whiteboard talk, *Columbia University*, New York, NY *Sep 2026*
- Contributed talk, ACES meeting (remote) *Jul 2026*
- **Invited seminar**, *Yale University*, New Haven, CT *Oct 2025*
- **Invited** TUNA Talk, *NRAO*, Charlottesville, VA *Oct 2025*
- Contributed talk, Stellar Origins, Austria *Sep 2025*
- **Invited seminar**, *ESO*, Germany *Sep 2025*
- Contributed talk, ALMA-IMF workshop (remote) *Jun 2025*
- Contributed talk, STScI Symposium, Baltimore, MD *May 2025*
- **Invited colloquium**, *University of Connecticut*, Storrs, CT *Apr 2025*
- Contributed talk, ACES meeting, Boston, MA *Aug 2024*
- **Invited seminar**, *University of Kansas* (remote) *Mar 2024*
- Contributed talk, Surveying the Milky Way, Pasadena, CA *Oct 2023*

Awards

ALMA Cycle 13; Band 10, C10 — **PI (8.1 hrs)**. *Disk substructures in Sgr B2 YSOs at 50 au resolution* *2026*
ALMA Ambassador, Cycle 13, **$2,500**. *Hosted ALMA workshops at UF and Columbia* *2026*
Various Travel Grants, **$3,200** *2025*

JWST Cycle 3 GO 5365 — **Co-PI (17.9 hrs)**. *Identifying YSOs in Sgr B2* *2024*
ALMA Cycle 11; Band 1, C10 — **PI (6.9 hrs)**. *Characterizing dusty YSOs and HII regions in Sgr B2* *2024*
Certificate of Outstanding Merit, UF International Center *2023*
Symposium Talk Award, UF Astronomy, **$100** *2023*
Distinguished Service & Citizenship, UF Astronomy, **$100** *2022*
NRAO Student Observing Support, ALMA Cycle 8, **$34,955** *2021*
Undergraduate Research Award, BU Astronomy *2020*

Workshops

Planets in the Galactic Context, New York, NY *2026*
Code/Astro, Evanston, IL *2024*
19th Synthesis Imaging Workshop, Charlottesville, VA *2023*
Star Formation School, Granada, Spain *2021*
GBT Remote Observer Certification, Green Bank Observatory, WV *2021*
Single Dish School, Green Bank Observatory, WV *2021*

Teaching & Mentoring

Project Advisor *Fall 2024 – present*
UF Star Formation Group
- Advising undergraduate (A. Daley) on a YSO catalog in Sgr B2 with ALMA. Student received $1,750 UF University Scholars award. A publication is on track to be submitted in Fall 2026.

Graduate Student Instructor *2020 – 2021*
Astronomy Laboratory, UF
- Taught undergraduate lab course; redesigned outdated experiments; produced remote-learning videos.

Teaching Assistant *Spring 2020*
Stellar & Galactic Astrophysics, Boston University
- Assisted students in an IDL-heavy upper-level astrophysics course.

Teaching Assistant *2018 – 2020*
Project Accelerate, Boston University
- Led lab sections of an Advanced Placement physics course for underserved students; developed review materials for each course module.

Science Communication & Community Engagement

- **Referee**, AAS Journals *2025 – present*
- **Webinar Author**, AstroSandbox — wrote and delivered 6 astronomy webinars for Ukrainian students *Python in astronomy*, *Astronomical image processing*, *Multiwavelength astronomy*, *Radio astronomy*, *How to write a telescope proposal?*, *Imposter syndrome*. *2020 – present*
- **Competition Judge**, AstroSandbox — judged annual astronomy team competition with 50 participants from 15 regions of Ukraine. *2020 – 2023*
- **Graduate Student Representative**, Mentoring Committee, UF Astronomy Department — gathered community feedback, synthesized findings into anonymized reports, and presented recommendations to faculty; served as liaison between student body and departmental leadership *2021 – 2023*
- **Mentor**, Ukraine Global Scholars, Ukraine Achievement Fund — guided 4 high-school students through university applications; 2 admitted to top-20 US boarding schools *2020 – 2024*
- **Mentor**, UF Women in Astrophysics & Astronomy Mentorship Program *2022 – 2024*
- **Pen Pal**, Letters to a Pre-Scientist. Year-long physical mail exchange with a high-school student *2021 – 2022*

Media Coverage

- UF press release: *UF researchers help capture most detailed view of the Milky Way's core* *2026*
- NASA press release: *NASA's Webb Explores Largest Star-Forming Cloud in Milky Way* *2025*
- UF press release: *UF astronomers use James Webb Space Telescope to uncover hidden stars in the Milky Way's largest stellar nursery* *2025*

Publications (full list: ADS library)

First Author

1. **Nazar Budaiev**, A. Ginsburg, A. T. Barnes, D. Jeff et al., *"JWST's first view of the most vigorously star-forming cloud in the Galactic center — Sagittarius B2."* AJ 172, 51 — *2026*
2. **Nazar Budaiev**, A. Ginsburg, C. Goddi, Á. Sánchez-Monge, A. Schmiedeke et al., *"Properties of H_2O masers and their associated sources in Sagittarius B2."* ApJ 989, 52 — *2025*
3. **Nazar Budaiev**, A. Ginsburg, D. Jeff, C. Goddi, F. Meng, Á. Sánchez-Monge et al., *"Protostellar cores in Sagittarius B2 N and M."* ApJ 961, 4 — *2024*

Significant Contribution

4. J. Wallace, C. Battersby, **Nazar Budaiev**, H. P. Hatchfield, A. T. Barnes et al., *"ACES VII: Compact continuum source catalog of the Central Molecular Zone."* ApJ 1009, 130 — *2026*
5. T. Yoo, A. Ginsburg, **Nazar Budaiev**, R. Galván-Madrid, A. Dawson et al., *"A JWST NIRCam/MIRI view of the W51A high-mass star-forming region."* AJ 171, 208 — *2026*
6. T. Yoo, A. Ginsburg, J. Braine, **Nazar Budaiev**, F. Louvet et al., *"ALMA-IMF XX: Core fragmentation in the W51 high-mass star-forming region."* ApJ 994, 233 — *2025*
7. S. Zhang, X. Lu, A. Ginsburg, **Nazar Budaiev**, Y. Cheng et al., *"Subclustering and star formation efficiency in three protoclusters in the Central Molecular Zone."* ApJ 982, L10 — *2025*

Contributing Author

8. A. Ginsburg, S. R. Gramze, M. L. N. Ashby, B. A. L. Gaches, **Nazar Budaiev** et al., *"The colors of ices: Measuring ice column density through photometry."* OJAp 963469G — *2026*
9. S. Gramze, A. Ginsburg, **Nazar Budaiev**, A. Bulatek, T. Richardson et al., *"Mapping CO ice in a star-forming filament in the 3 kpc Arm with JWST."* OJAp 965431G — *2026*
10. F. Xu, X. Lu, K. Wang, H. B. Liu (incl. **Nazar Budaiev**), *"DUET: Cloud-wide census of continuum sources showing low spectral indices."* A&A 697, A164 — *2025*
11. A. Ginsburg, J. Bally, A. T. Barnes, C. Battersby, **Nazar Budaiev**, et al., *"A broad line-width, compact, millimeter-bright molecular emission line source near the Galactic center."* ApJ 968, L11 — *2024*
12. D. Jeff, A. Ginsburg, A. Bulatek, **Nazar Budaiev**, Á. Sánchez-Monge et al., *"Thermal properties of the hot core population in Sagittarius B2 Deep South."* ApJ 962, 48 — *2024*
13. Y. T. Yan, C. Henkel, K. M. Menten, Y. Gong, H. Nguyen (incl. **Nazar Budaiev**), *"Discovery of non-metastable ammonia masers in Sagittarius B2."* A&A 666, L15 — *2022*
14. F. Meng, Á. Sánchez-Monge, P. Schilke, A. Ginsburg, C. DePree (incl. **Nazar Budaiev**), *"The physical and chemical structure of Sagittarius B2. VI. UCH*II *regions."* A&A 666, A31 — *2022*

Contributing Author — ACES Large Program (incl. data reduction)

15. Y. Sofue, S. N. Longmore, D. Walker, A. Ginsburg, J. D. Henshaw (incl. **Nazar Budaiev**), *"Electron temperature and emission measure of H*II *regions in the central molecular zone (CMZ) from H40α recombination line and continuum emissions by ALMA CMZ Exploration Survey (ACES)."* PASJ 78, 1954 — *2026*
16. D. R. Lipman, J. Poznanski, S. R. Gramze, C. E. Cook, C. Battersby (incl. **Nazar Budaiev**), *"ACES VIII: A survey of compact, high-velocity features observed in CS(2–1)."* Accepted to ApJ arXiv:2609.10472 — *2026*
17. S. N. Longmore, J. Bally, A. T. Barnes, C. Battersby, L. Colzi, A. Ginsburg (incl. **Nazar Budaiev**), *"ACES I: Overview."* MNRAS 552, 377 — *2026*
18. A. Ginsburg, D. L. Walker, A. T. Barnes, X. Lu, Á. Sánchez-Monge, J. E. Pineda (incl. **Nazar Budaiev**), *"ACES II: 3 mm continuum images."* MNRAS 552, 354 — *2026*
19. D. L. Walker, A. Ginsburg, A. T. Barnes, X. Lu, P.-Y. Hsieh, Á. Sánchez-Monge, S. R. Gramze, **Nazar Budaiev** et al., *"ACES III: Molecular line data reduction and HNCO and HCO^+ data."* MNRAS 552, 256 — *2026*
20. X. Lu, D. L. Walker, A. Ginsburg, A. T. Barnes, P.-Y. Hsieh, Á. Sánchez-Monge, S. R. Gramze, **Nazar Budaiev** et al., *"ACES IV: Data of the two intermediate-width spectral windows."* MNRAS 552, 255 — *2026*
21. P.-Y. Hsieh, D. L. Walker, A. Ginsburg, A. T. Barnes, X. Lu, Á. Sánchez-Monge, S. R. Gramze, **Nazar Budaiev** et al., *"ACES V: CS(2–1), SO(2_3–1_2), CH_3CHO($5_{1,4}$–$4_{1,3}$), HC_3N(11–10), and H40α lines data."* MNRAS 552, 253 — *2026*